

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built an **AI-assisted workflow** for company-registration drafts, combining structured prompt outputs, operator validation, public-site API lookups, and automation handoff.
- Implemented async backend execution with retries, session recovery, browser automation, and state validation, reducing manual filing time by **85%**.
- Containerized services with Docker, deployed to Google Cloud Run, and automated releases through GitHub Actions, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built Python medical-image segmentation tooling for **95 brain subparts**, covering preprocessing, orientation correction, model inference, generated masks, and large-volume MRI data paths.
- Reduced segmentation runtime by **87%** using asynchronous and parallel execution while preserving inspection workflows for generated masks and 3D surfaces.
- Debugged model-output pipeline issues across conformation, label mapping, PyTorch/ONNX-style inference, and binary mask generation inside a mature open-source codebase.

Open Source

[Contributions](#) 

- Contributed to **Invesalio** model-output and PyTorch/ONNX-adjacent workflows, plus **IOOS/Tiled** data and visualization tooling and VideoLAN/CRIU parsing or systems robustness fixes under maintainer review.

Projects

Swish Support System | *Python, FastAPI, PostgreSQL/pgvector, Redis, Langfuse, React*

[GitHub](#) 

- Built retrieval-backed support workflows where LLM calls use customer, order, item, evidence, and policy context before producing traceable assessments and responses.
- Separated semantic understanding from deterministic action rules, adding confidence thresholds, human handoff paths, Langfuse traces, and regression checks for ambiguous conversations.

Penny Lane Capital | *Python, Agno, Pydantic, SQLite, Vector Memory, JSONL*

[GitHub](#) 

- Built a multi-agent research workflow with analyst roles, bull/bear debate, trader, risk review, portfolio approval, typed Pydantic state, SQLite checkpoints, and JSONL traces.
- Implemented **vector memory** and semantic lesson retrieval so agents could pull similar past decisions into context and inspect retrieval quality through validation harnesses.

Hyoka | *Python, FastAPI, OpenTelemetry, PostgreSQL, Typer, Next.js, Docker*

- Built a pluggable **AI-agent evaluation and replay layer** that captures LLM, tool, retrieval, memory, latency, cost, error, and metadata traces through SDK, proxy, event, scrape, and OTLP/JSON ingestion paths.
- Added evaluator registries, replay modes, failure mining, evidence-hashed candidate proposals, validation runs, release gates, signed manifests, and worker leases for controlled agent-behavior promotion.

Technical Skills

Languages: Python, TypeScript, Go, C++

Frameworks: PyTorch, FastAPI, Node.js, React, Next.js, Agno

Tools: PostgreSQL/pgvector, Redis, SQLite, Vector Memory, Langfuse, OpenTelemetry, Docker, GCP, GitHub Actions, ONNX